Numerical Methods/Analysis

- Use of mathematical software packages
- From Models to Analytical and to Numerical Simulation
- Example

Motivation

- Science and engineering demand a quantitative analysis of physical phenomena. Such an analysis requires a sophisticated mathematical apparatus.
- Computers are very helpful; several software packages for mathematical software exist.
  - Specialized packages such as Ellpack for solving elliptic boundary value problems.
  - General-purpose systems are:
    - (i) Mathematica of Wolfram Research;
    - (ii) Maple of Maplesoft;
    - (iii) Matlab of Mathworks); and
    - (iv) IDL numeracy

Mathematica

- All-purpose mathematical software package.
- It integrates
  - swift and accurate symbolic and numerical calculation,
  - all-purpose graphics, and
  - a powerful programming language.
- It has a sophisticated “notebook interface” for documenting and displaying work. It can save individual graphics in several graphics format.
- Its functional programming language (as opposed to procedural) makes it possible to do complex programming using very short concise commands;
- It does, however, allow the use of basic procedural programming constructs like Do and For.
- Drawbacks: steeper learning curve for beginners used to procedural languages; more expensive.

Maple

- Powerful analytical and mathematical software.
- Does the same sorts of things that Mathematica does, with similar high quality.
- Maple's programming language is procedural (like C or Fortran or Basic) although it has a few functional programming constructs.
- Drawbacks: Worksheet interface/typesetting not as developed as Mathematica's, but it is less expensive.

Matlab

- Combines efficient computation, visualization and programming for linear-algebraic technical work and other mathematical areas.
- Widely used in the Engineering schools.
- Drawbacks: Does not support analytical/symbolic math.

Models

- Prototype of real world
- Abstractions of physical, social, economical, systems or phenomena.
- Design to allow us to understand complex systems or phenomena.
- A model captures <u>only</u> aspects of the original system relevant for the type of analysis being conducted.
- Example: the study of the liftoff properties of a wing in a wind tunnel.

Computer simulation

- The process of imitating a real phenomenon with a set of mathematical formulas (model)
- Theoretical studies, experiment and computer simulation are three investigative methods in science and engineering.
- In this class we are only concerned with computer models of physical systems.

Mathematical Models

- A formulation or equation that expresses the essential features of a physical system or process in mathematical terms.
- Models can be represented by a functional relationship between:
  - dependent variables,
  - independent variables,
  - parameters, and
  - forcing functions.

$$\textit{dependent variable} = f(\textit{independent variable}, \textit{parameters}, \textit{forcing function})$$

Mathematical Model (cont'd)

- *Dependent variable* → a characteristic that usually reflects the behavior or state of the system
- *Independent variables* → dimensions, such as time and space, along which the system's behavior is being determined
- *Parameters* → constants reflective of the system's properties or composition
- *Forcing functions* → external influences acting upon the system

Mathematical Model (cont'd)

- Conservation laws provide the foundation for many model functions. Examples of such laws:
  - Conservation of mass
  - Conservation of momentum
  - Conservation of charge
  - Conservation of energy

- Some system models will be given as implicit functions or as differential equations - these can be solved either using analytical methods or numerical methods.

Analytical versus numerical methods for model solving

- Once a mathematical model is constructed one could use
  - Analytical methods
  - Numerical methods
- Analytical methods
  - Produce exact solutions
  - Not always feasible
  - May require mathematical sophistication
- Numerical methods
  - Produce an approximate solution
  - The time to solve a numerical problem is a function of the desired accuracy of the approximation.

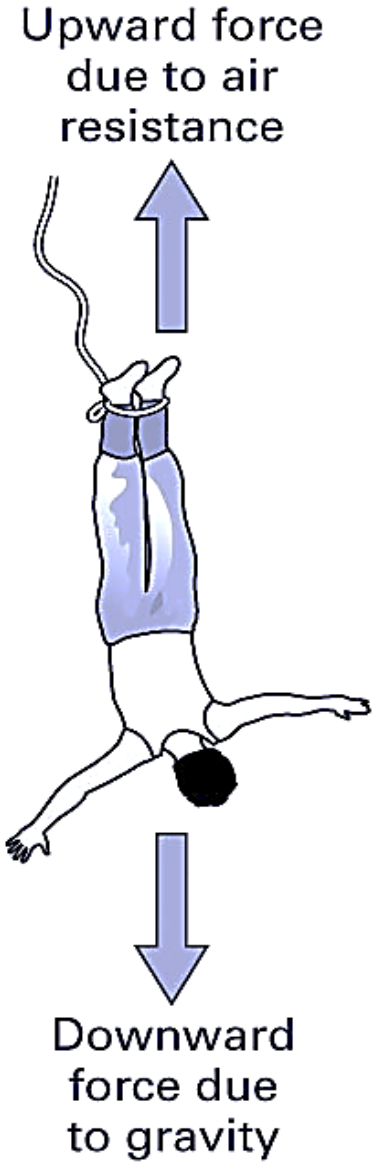
Example: the analytical model

Consider a bungee jumper in midair.
The model for its velocity is given by the differential equation:

$$\frac{dv}{dt} = g - \frac{c_d}{m} v^2 \qquad \text{Equation (1)}$$

The change in velocity is affected by: the gravitational force which pulls it down and are opposed by the drag force

Dependent variable – downward velocity $\boldsymbol{v}$ in m/s
Independent variables - time t in s
Parameters – jumper's mass m *in kg*,
Jumpers drag coefficient c_d in kg/m
Forcing function - gravitational acceleration g in m/s^2



Example: the analytical model

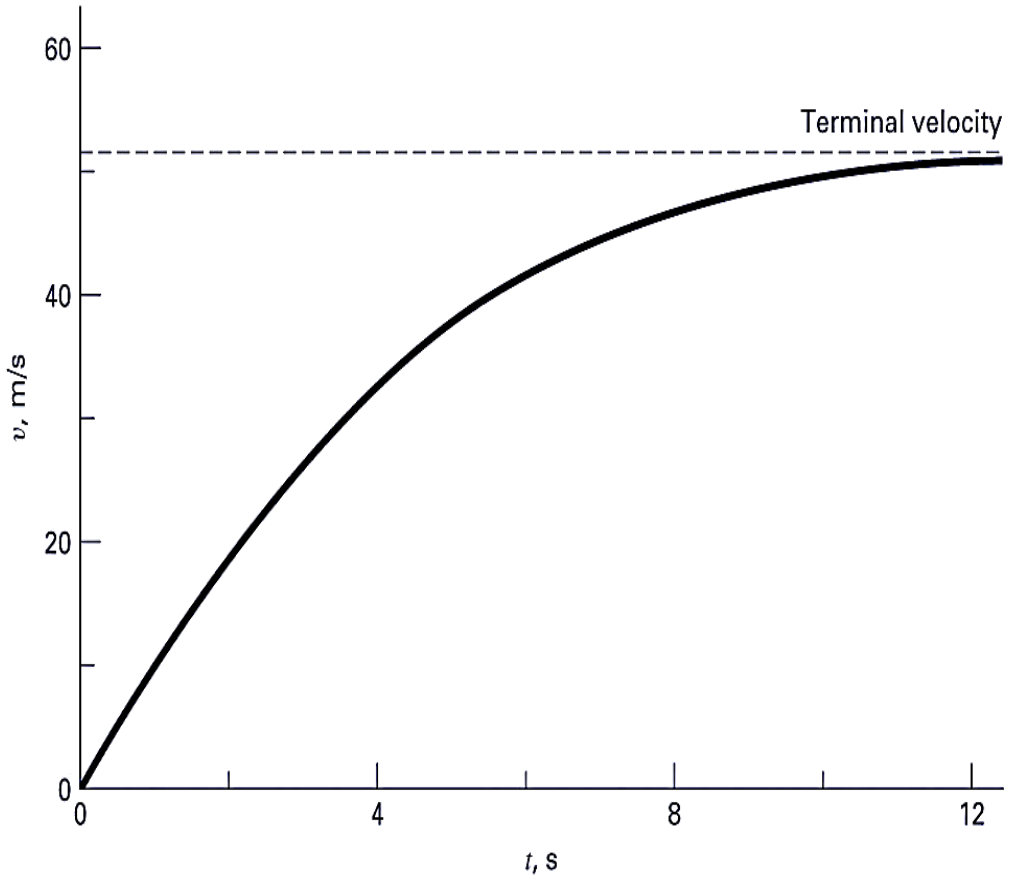
- To understand eqn.(1)
- Newtons 2nd Law, $F = ma$ or $a = F/m = dv/dt$
- If the net force is positive, the object will accelerate,
- Acceleration = rate of change of velocity with respect to time.
- The net force $F = F_D + F_U$
- Downward pull of gravity $F_D = mg$
- Upward force of air resistance $F_U = -c_d v^2$

- $dv/dt = F/m = F_D/m + F_U/m \Rightarrow \frac{dv}{dt} = g - \frac{c_d}{m} v^2$

Example – the analytical solution

$$v(t) = \sqrt{\frac{gm}{C_d}} \tanh\left(\sqrt{\frac{gC_d}{m}}\, t\right)$$

The model can be used to generate a graph. Example: the velocity of a 68.1 kg jumper, assuming a drag coefficient of 0.25 kg/m



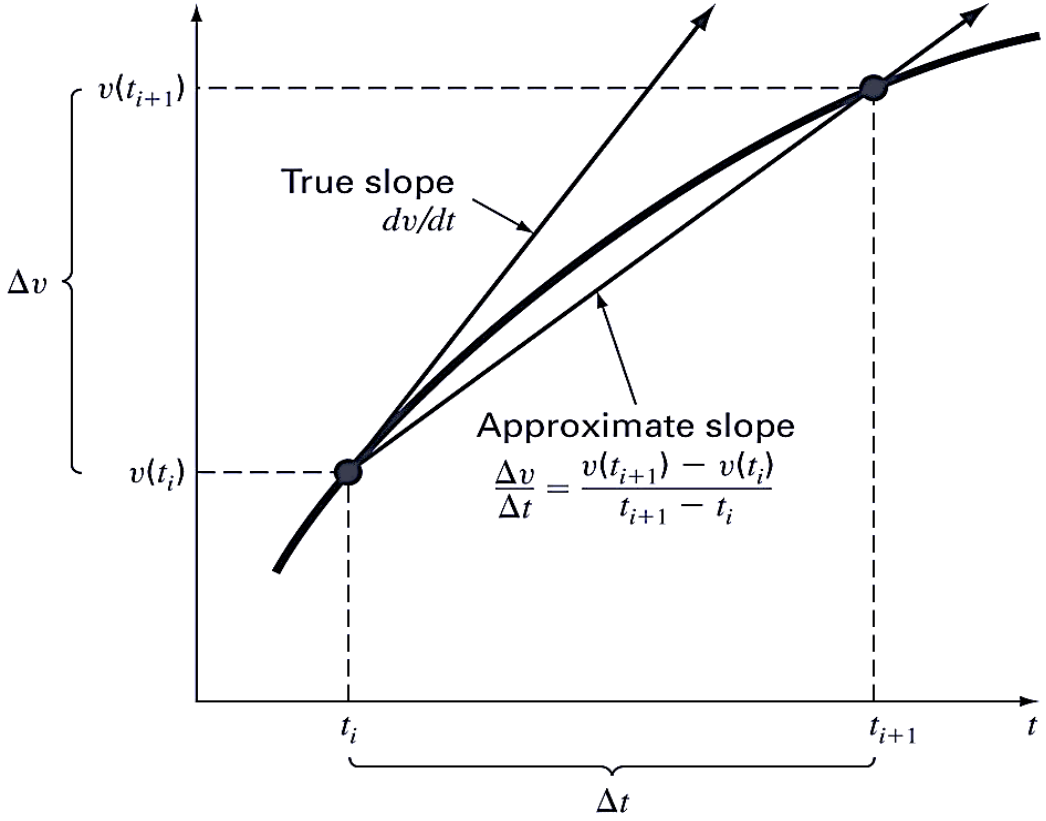
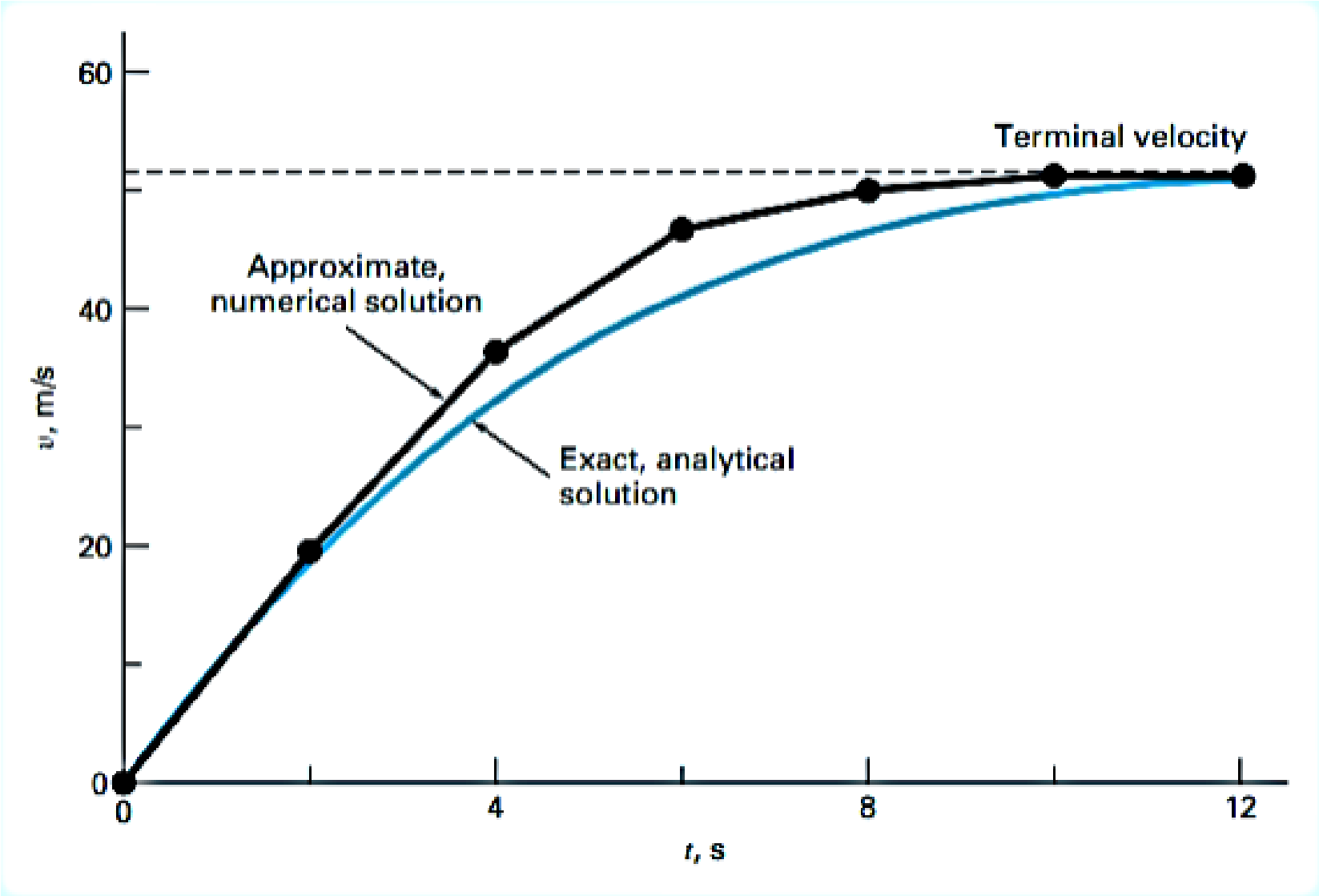
Example: numerical solution

- For the numerical solution we observe that the time rate of change of velocity can be approximated as:

$$\frac{dv}{dt} \approx \frac{\Delta v}{\Delta t} = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i}$$

$$dv_i/d_t = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i} = g - \frac{c_d}{m} v(t_i)^2$$

$$v(t_{i+1}) = v(t_i) + [g - \frac{c_d}{m} v(t_i)^2](t_{i+1} - t_i)$$



Example: numerical solution